Juan Pablo Garcia Chavez

Junior Machine Learning | Python | TensorFlow | Scikit-learn | Pipelines

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Profile

Junior Machine Learning practitioner with a background in Mechatronics Engineering and a Master's degree in Electronic Engineering. I work with classification, regression, computer vision, NLP, forecasting, and signal-focused projects. My current focus is building reproducible solutions and moving from notebooks toward reusable workflows.

Skills

- Python, scikit-learn, TensorFlow, Keras, LightGBM.
- Training, validation, and evaluation pipelines.
- Computer vision, NLP, forecasting, and signal processing.
- Trade-off analysis between predictive quality, speed, and complexity.

Selected Projects

Electrical disturbance classification | Applied project

- Context: classified electrical disturbances to support power-quality analysis.
- Technologies: Python, NumPy, pandas, scikit-learn, TensorFlow, and signal processing.
- Measurable result: reported accuracy above 96% and roughly 80% processing reduction.
- Impact: connected engineering background with ML applied to signals and operational efficiency.

Age estimation with computer vision | Applied academic project

- Context: estimated age from images as operational support in a sensitive use case.
- Technologies: Python, TensorFlow, Keras, ResNet50, and GPU training.
- Measurable result: 7.03 validation MAE with ResNet50 and a regression head.
- Impact: paired deep learning with responsible judgment: review support, not a single automated decision.

IMDB review sentiment | Applied academic project

- Context: classified positive and negative reviews to monitor user perception.
- Technologies: Python, TF-IDF, logistic regression, LightGBM, and a controlled BERT test.
- Measurable result: F1 >= 0.85 target reached with a TF-IDF plus linear-model approach.
- Impact: showed that an efficient, explainable solution can be competitive before moving to heavier models.

Taxi order forecasting | Applied academic project

- Context: forecasted hourly demand to anticipate peak-hour operations.
- Technologies: Python, pandas, scikit-learn, lags, rolling means, and Random Forest.
- Measurable result: 43.21 test RMSE, meeting the operational target.
- Impact: built a reproducible temporal pipeline with chronological validation.

Education

- Master's degree in Electronic Engineering.
- Mechatronics Engineering.
- Applied training in Data Science and Machine Learning.

Links

- Machine Learning dashboard: https://JuanPa7799.github.io/juanpablo-data-portfolio-/dashboards/machine-learning/
- Main portfolio: https://JuanPa7799.github.io/juanpablo-data-portfolio-/